BRANDON LUU

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EDUCATION

B.S. Mathematics-Computer Science, University of California, San Diego June 2026
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Software Engineering, Machine Learning, Neural Networks, Deep Learning, Advanced Machine Learning Methods

SKILLS

Languages	Python, Go, TypeScript, JavaScript, C++, Java, SQL
Backend & Data	Data Pipelines, Event-Driven Architecture, REST APIs, Microservices, PostgreSQL, SQLite
Reliability	Fault Tolerance, Idempotency, Recovery/Retries, Monitoring, Observability, Alerting
Distributed Systems	Distributed Systems, Concurrency, Caching, Scalability, High-Throughput Processing
Performance	Performance Profiling & Optimization, Latency Analysis, GPU/CUDA, Nsight Compute
Cloud & DevOps	AWS (Lambda, SQS, API Gateway), Docker, Linux, Git, CI/CD, Testing & Automation
Foundations	Data Structures & Algorithms, System Design, Object-Oriented Design, Debugging, Code Review
AI & Tooling	LLMs, RAG, Agent Workflows; Claude Code, Cursor

EXPERIENCE

Software Engineer Intern, Mosa Tea Dec 2025 – July 2026, San Diego, CA

- Built and deployed a real-time data pipeline (API Gateway → SQS → Lambda) in Python on Linux, ingesting 150+ daily event streams into structured, analysis-ready data and owning it in production after launch.
- Engineered a fault-tolerant, idempotent processing layer with state tracking and upsert-based deduplication that handled late and duplicate events, eliminating duplicate-processing errors and preserving data integrity across async retries.
- Integrated multiple external APIs (Square Loyalty, Customers, Orders, Payments) into consolidated data views spanning 5,000+ users for operational analysis.
- Wrote automation scripts, unit tests, and validation harnesses to verify pipeline behavior and surface regressions early.
- Used AI coding assistants (Cursor, Claude Code) to move faster, reviewing and understanding all generated code before merging to production.

PROJECTS

- **MetricFlow (Real-Time Metrics & Monitoring Engine)** Go, Concurrency, System Design
  Built a metrics ingestion and aggregation engine from scratch in Go (net/http) that surfaces what a running system is doing: HTTP ingest/query endpoints with live per-metric aggregates over a rolling 60s window, using fixed-width time buckets to keep memory and query cost flat regardless of event volume, with mutex-serialized concurrent writes.
- **Postmortem Generator** Python, FastAPI, LangGraph, LangChain
  Built an agentic incident-analysis tool using a 6-node LangGraph workflow to parse raw logs, identify root cause, and generate structured postmortem reports, served via an async FastAPI REST API with full LangSmith observability.
- **EmptyBoard (Classroom Availability Engine)** TypeScript, React, Node.js, REST APIs
  Shipped a real-time classroom-availability web app, from a React/TypeScript frontend to a Node backend, over 10,000+ UCSD sections, returning open-room results in under 200ms via time-window filtering and live schedule parsing.
- **CUDA SGEMM Optimization** C++, CUDA, Nsight Compute
  Wrote a single-precision matrix-multiply kernel from scratch and tuned it through a coalesced, shared-memory, register-blocked, vectorized, and warp-tiled kernel ladder to reach ~8,200 GFLOPS (~95% of cuBLAS, 1024^3) on an RTX 4060, profiling each bottleneck with Nsight Compute.
- **Mosa RAG System** Python, LLMs, Vector Embeddings, Semantic Search
  Built a retrieval-augmented generation pipeline over 130+ operational records, reaching 100% Hit@5 and MRR 1.00 across a 39-query evaluation set; accelerated inference with ONNX embeddings for sub-minute lookups.